Knowing When the Evidence Is Not Enough: Reinforcement Learning for Calibrated Fact Verification

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Background

Platforms are handing fact-checking to AI. Meta ended its US fact-checking programme in April 2025 [1], and 447,083 posts on X asked Grok for a fact check in six months of 2025, yet the bots agreed with professional fact-checkers only about half the time [2]. A fact-checker that answers confidently when the evidence is unclear spreads false confidence. Datasets such as FEVER [3], SciFact [4] and AVeriTeC [5] have answer keys, so verdicts can be scored automatically, which reinforcement learning (RL) needs. RL has been used to make fact-checkers cite evidence [6] and check claims in steps [7], and to teach honest confidence and "I don't know" in question answering [8, 9]. None do this for fact-checking.

Problem statement

Given a claim, the model must say whether it is Supported, Refuted or Not Enough Info (NEI), point to the Wikipedia sentences it relied on, and state how confident it is. Small models that fit on one GPU do this poorly. On 900 FEVER claims, Qwen3.5-2B is right only 63% of the time even when shown the correct evidence (a 180B model: 95%). It calls almost everything Supported and almost always claims 100% confidence. Can RL teach a small model to weigh evidence, say NEI when the evidence falls short, give a confidence that matches its accuracy, and treat calling a false claim true as the worst mistake?

Approach

Verifier: fine-tune the model on labelled examples (SFT), then train it further with RL (GRPO [10]). The reward pays for correct verdicts, penalises calling false claims true the most, rewards citing the right sentences, and scores stated confidence against actual accuracy. Varying the NEI reward shows how accuracy trades off against how often the model abstains, so we can choose how cautious it is. **Controller:** a small second model that decides whether to search again, split the claim, verify, abstain or answer, with a cost for each search. **Evaluation:** one shared script scores every run for accuracy, confidence calibration and search cost, and compares RL against the SFT model and simple baselines [11]. We test for shortcuts with the FEVER symmetric set [12] and by swapping evidence between claims, and test on SciFact and AVeriTeC to see if the model generalises.

Team roles and deadlines

	Agentic group (AG-1, AG-2, AG-3)	RL group (RL-1, RL-2, RL-3)
Wk 7	Agent loop, dataset loaders, shortcut tests	Try the 4B model, fix the reward
Wk 8–9	Agent baseline, multi-step RL training	RL training, reward experiments
Wk 10–11	Tests on new datasets, error analysis, demo	Controller training, cost experiments
Wk 12	Final report, slides and live demo	

Risk assessment and preliminary results

We have already tested the riskiest steps. Fine-tuning took 37 minutes on our 16 GB GPU (RTX 5070Ti).

- **The 2B model is too small.** Fine-tuning raised it from 63% to 82% with the correct evidence shown. In week 7 we try the 4B model (about 1.5 hours as a test) before choosing.
- **RL adds nothing over fine-tuning.** Fine-tuning alone raised correct NEI answers from 14% to 72% (180B model: 35%). RL targets what fine-tuning cannot fix: the model still always claims 100% confidence, and says NEI on 20% of false claims even when shown evidence that disproves them.
- **The model takes shortcuts.** Given the wrong claim's evidence, it says NEI on 98% of true claims, so it does read the evidence. However, it still calls 23% of false claims false from their wording alone, so both shortcut tests run on every training run.
- **The model says NEI to everything to play safe.** We report accuracy at every level of caution.
- **It only works on FEVER.** We test on AVeriTeC's real-world claims and report the gap.
- **One GPU for two training jobs.** We share GPU time, and the controller trains on CPU.

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